

Figure 1

[FIGURE 1 — header image, placed above or directly under the title] **Caption:** *The game: Alice sees the whole battlefield, Bob operates the gun, and neither controls where it points. One bit connects them.* **Alt-text:** An 8-by-8 naval battle grid with ships hidden in some cells. On one side a figure (Alice) in a watchtower overlooking the full grid; on the other a figure (Bob) at a gun aimed at a single highlighted cell. A single line between them carries the message "1 bit".

Alice

★	★		★
★		★	
	★	★	★
★		★	



Alice knows the location of the enemy ships (and Bob does not).

One bit communication



Bob

0	1	2	3
4		6	7
8	9	10	11
12	13	14	15



Bob knows where the gun is pointing (and Alice does not).

Bob to decide:

Shoot

No shoot

Rules

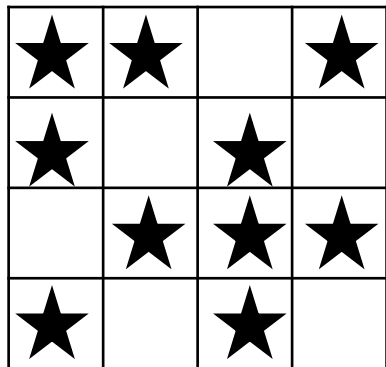
1. Game is played on $n \times n$ grid
2. Each cell has equal probability to contain enemy ship
3. Alice knows the position of enemy ships
4. Bob knows where the gun is pointing
5. Alice communicates 1 bit to Bob
6. Bob decides whether to shoot, or not shoot the gun
7. If they hit the enemy or decide to not shoot an empty cell they win.
8. If they shoot an empty cell or not shoot the enemy, they lose.

Winning is:

- Shoot the enemy
- Not shoot an empty cell

Figure 2

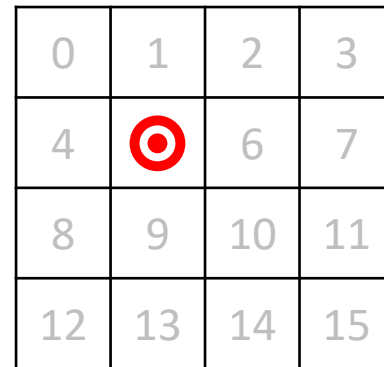
[FIGURE 2 — placed after the paragraph above] **Caption:** *Stripped of the ships: a 2×2 board read in a fixed order is a string of four bits — ship is 1, sea is 0. The gun's cell is Bob's index.* **Alt-text:** Left: a 2-by-2 battle grid with a ship in the top-left and bottom-right cells, cells numbered 1 to 4 in reading order. Right: the same board written as the bitstring 1001. An arrow labelled "same game" connects them.



Hidden binary field (only known to Alice)

Alice's bit string (representing the ships on the board)

1	1	0	1	1	0	1	0	0	1	1	1	1	0	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



Random query index (only known to Bob)

Bob's assignment (representing gun position)

index to guess: 5 (first cell is index 0, last index 15)

Bob guesses:

0 → Win!

1 → Lose

Figure 3

[FIGURE 3 — placed here, before the counting paragraph] Caption: *Alice's encoding for $n = 4$: all sixteen boards arranged by number of ships (1, 4, 6, 4, 1). The bottom rows say 'hold', the top rows say 'shoot' — and the six tied boards in the middle belong to neither until a tie-break agreement assigns them.* **Alt-text:** Sixteen small 2-by-2 boards arranged in five rows by ship count: one board with no ships, four with one ship, six with two ships, four with three ships, one with four ships. The bottom two rows are tinted blue and labelled "send: hold"; the top two rows are tinted red and labelled "send: shoot"; the middle row of six boards is grey and labelled "tied — no majority".

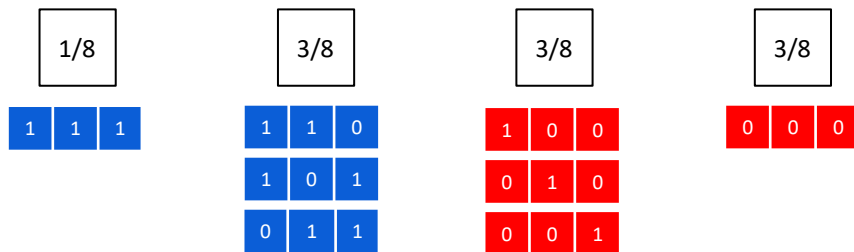
Majority strategy: Alice sends the majority value, and Bob guesses whatever Alice sends

1 1 1 1	}	4 ships. Alice sends 1. Bob success rate: 100%. Occurrence: 1/16
1 1 1 0	}	3 ships. Alice sends 1. Bob success rate: 75%. Occurrence: 4/16
1 1 0 1		
1 0 1 1		
0 1 1 1		
1 1 0 0	}	2 ships. Alice sends ?. Bob success rate: 50%. Occurrence: 6/16
1 0 1 0		
1 0 0 1		
0 1 1 0		
0 1 0 1		
0 0 1 1		
1 0 0 0	}	1 ship. Alice sends 0. Bob success rate: 75%. Occurrence: 4/16
0 1 0 0		
0 0 1 0		
0 0 0 1		
0 0 0 0	}	0 ship. Alice sends 0. Bob success rate: 100%. Occurrence: 1/16

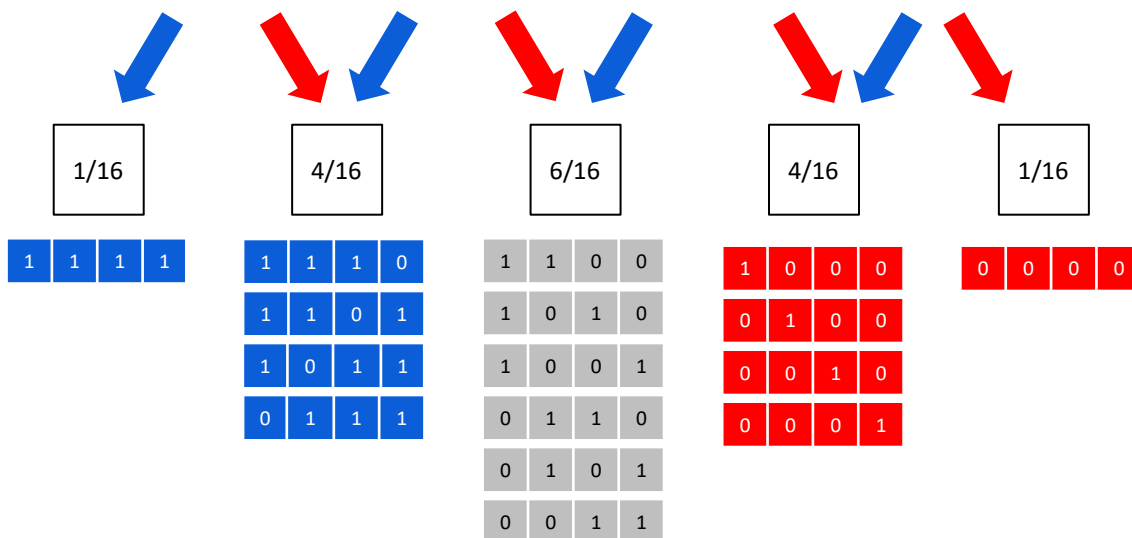
Figure 4

[FIGURE 4 — the table, rendered as an image for Medium; placed here] **Caption:** *Three tie-break agreements for $n = 4$. Every information measure falls down the ladder; the win rate does not move.* **Alt-text:** A table with three rows and four columns. Row 1: copy cell 1 — 1.000 bits about the board, 0.148 bits about Bob's cell, win rate 11/16. Row 2: always 'shoot' — 0.896 bits, 0.124 bits, win rate 11/16. Row 3: flip a coin — 0.625 bits, 0.104 bits, win rate 11/16. The win-rate column is highlighted to emphasise that it is identical in all three rows.

[FIGURE 5 — centrepiece; placed here] Caption: *The cancellation, drawn. Top: row $n - 1$ of Pascal's triangle for $n = 4(1, 3, 3, 1)$ and $n = 8(1, 7, 21, 35, 35, 21, 7, 1)$, coloured by outcome at Bob's cell; mirror pairs cancel win against loss, and the one cell left of the axis — the tie — survives. Bottom: the same rows normalised for $n = 4, 8, 16$, the bell curve emerging and the surviving cell shrinking like $1/\sqrt{n}$. * **Alt-text:** Top half: two horizontal bar rows. First row has four blocks sized 1, 3, 3, 1; second row has eight blocks sized 1, 7, 21, 35, 35, 21, 7, 1. In each, blocks right of the central dashed axis are green and labelled "win", blocks left are red and labelled "loss", except the block immediately left of the axis, which is yellow and labelled "tie — pays 1/2". Curved arcs pair each green block with its red mirror image. Bottom half: three normalised bar silhouettes for $n = 4, 8, 16$ with a bell curve overlaid, the yellow central block visibly smaller in each successive silhouette.



All bits other than the one Bob has to guess (8 options).



The bit Bob has to guess

Alice's bits (16 options)

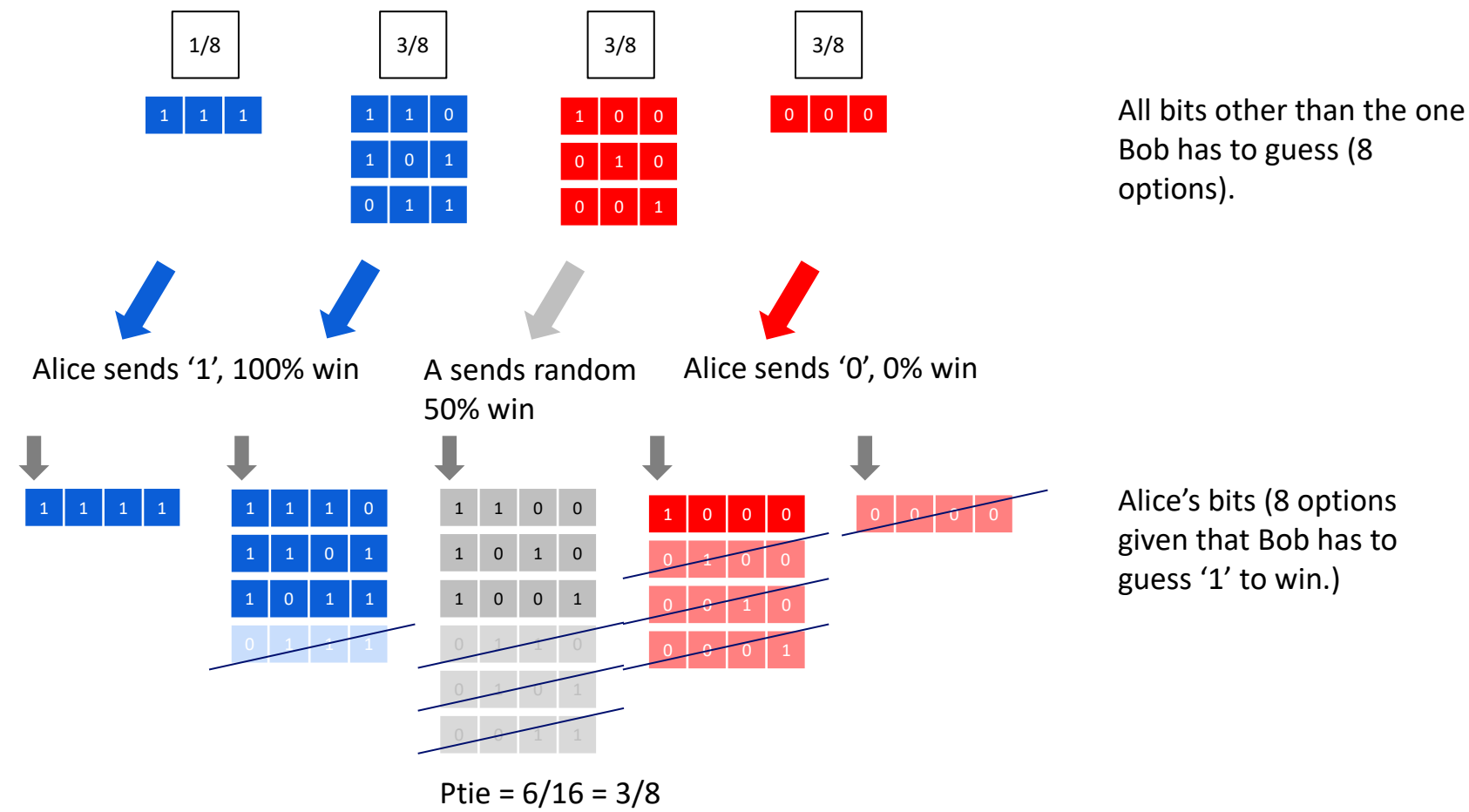
Alice communicates '1' if the majority is '1'

When string is 'tied' Alice sends a random '1' of '0'

Alice communicates '0' if the majority is '0'

Instead of asking how likely is Bob to guess right, we can ask the question: Given that Bob has to guess '1' in the first index, how likely is Alice to send this value?

There are 8 possible strings determined by the values of the other bits. If these 3 other bits have a majority of 1's Alice will send a 1 and they will win with 100% certainty. If the there is no '1' in the other three bits Alice will send a '0' and have 0% change of winning. Only if the there is just one '1' in the other three bits Alice will send a random bit and the win rate is 50%.

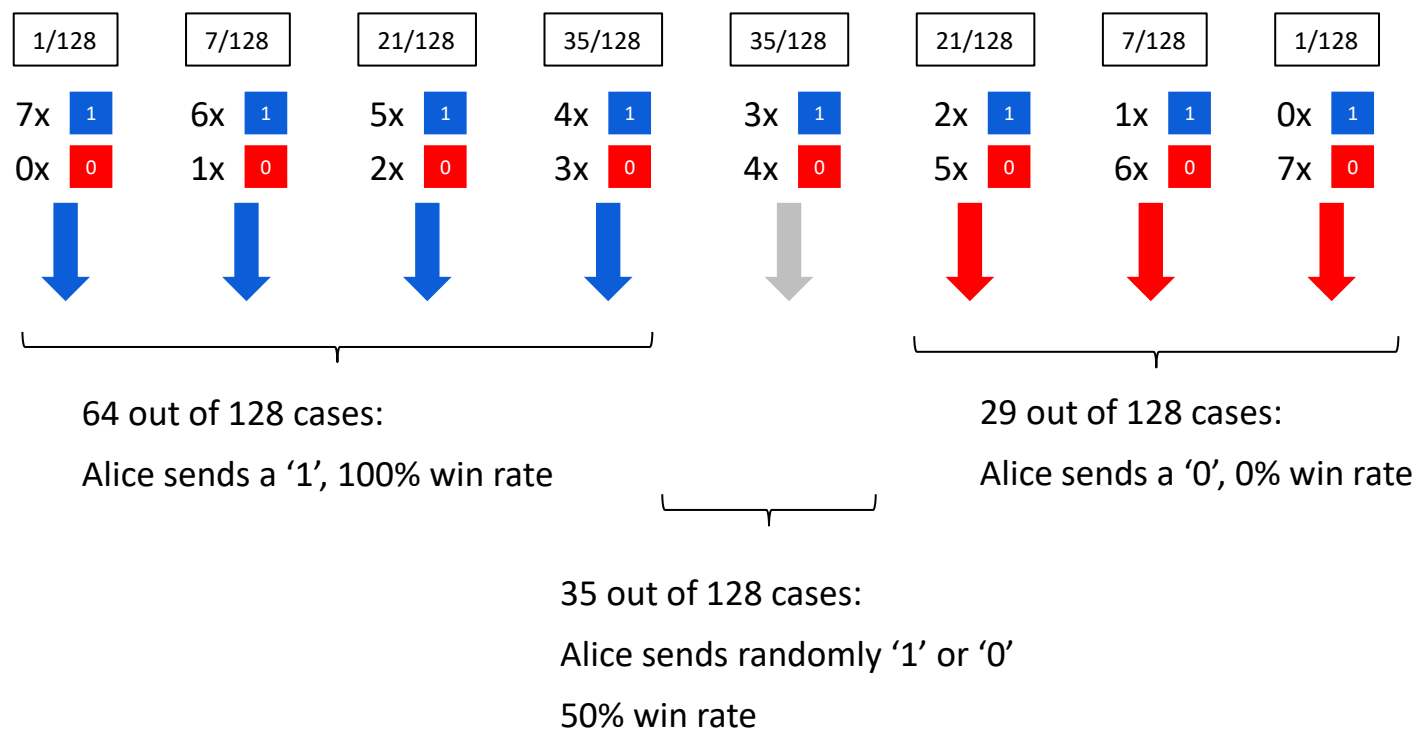


Win rate =
 $1 \times \frac{1}{8} + 1 \times \frac{3}{8} + \frac{1}{2} \times \frac{3}{8} = \frac{11}{16}$

Win rate =
 $\frac{1}{2} + \frac{1}{2} \text{ Ptie}$

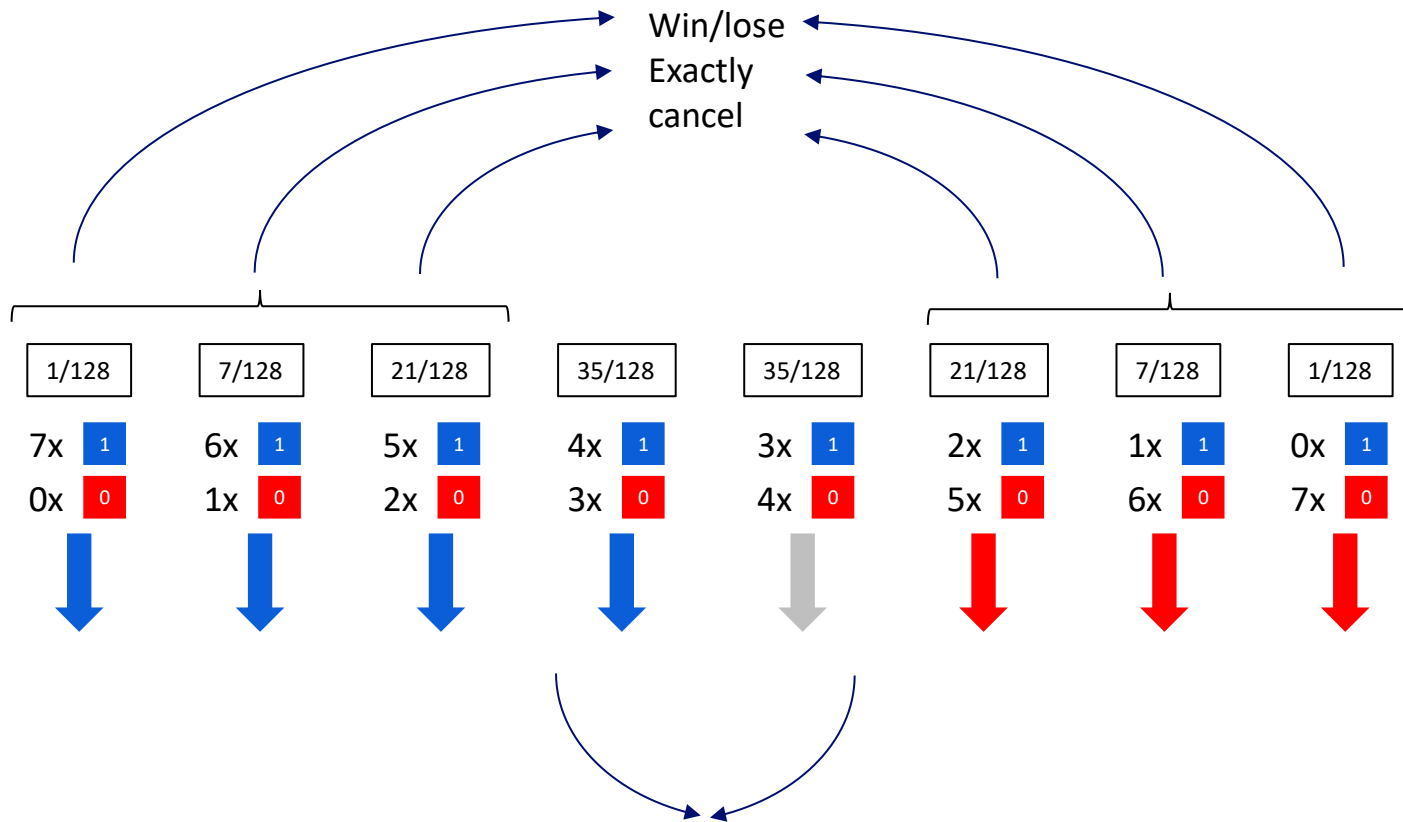
For $n = 8$, if Bob has to guess '1' on the first index. How likely is Alice to send a '1'

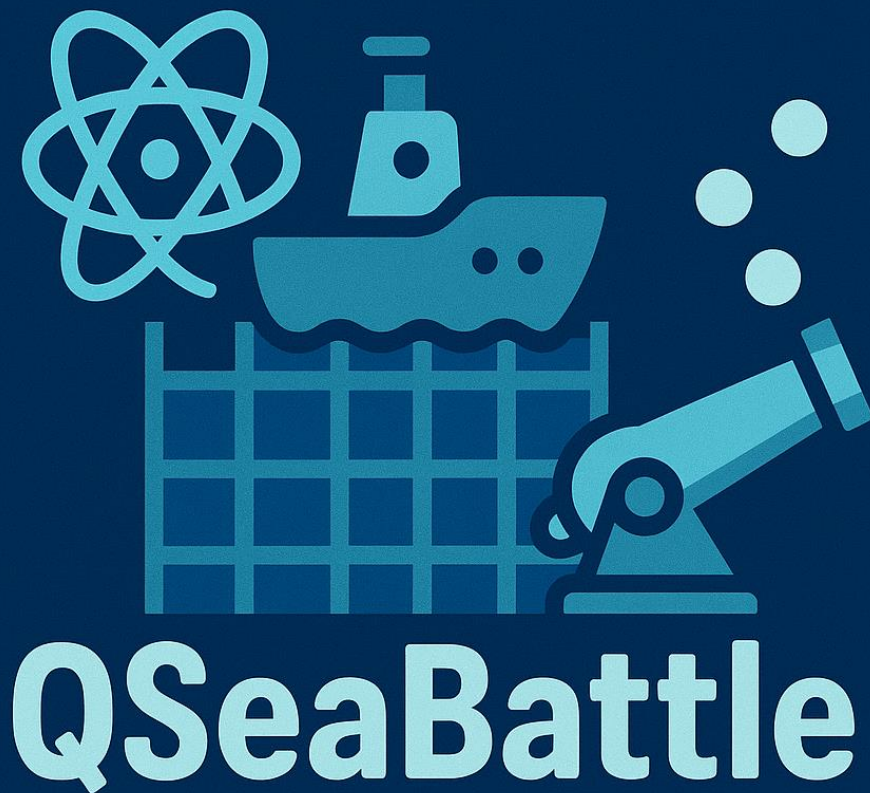
Evaluate the other 7 bits on their Hamming weight.



$$\text{Win rate} = 64/128 + 1/2 \times 35/128$$

$$\text{Win rate} = \frac{1}{2} + \frac{1}{2} \text{Ptie}$$





QSeaBattle

A two-player game where success is limited by information flow

How to benefit from (quantum) non-locality in an information game.

Rob Hendriks, January 2026